

Gates

Six types of gates

- NOT
- AND
- OR
- XOR
- NAND
- NOR

NOT Gate

A NOT gate accepts one input signal (0 or 1) and returns the complementary (opposite) signal as output

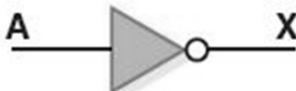
Boolean Expression	Logic Diagram Symbol	Truth Table						
$X = A'$		<table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	X	0	1	1	0
A	X							
0	1							
1	0							

FIGURE 4.1 Representations of a NOT gate

AND Gate

An AND gate accepts two input signals

If both are 1, the output is 1; otherwise, the output is 0

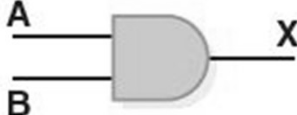
Boolean Expression	Logic Diagram Symbol	Truth Table															
$X = A \cdot B$		<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	X	0	0	0	0	1	0	1	0	0	1	1	1
A	B	X															
0	0	0															
0	1	0															
1	0	0															
1	1	1															

FIGURE 4.2 Representations of an AND gate

OR Gate

An OR gate accepts two input signals

If both are 0, the output is 0; otherwise, the output is 1

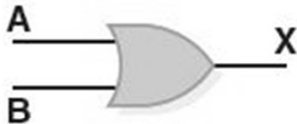
Boolean Expression	Logic Diagram Symbol	Truth Table															
$X = A + B$		<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	X	0	0	0	0	1	1	1	0	1	1	1	1
A	B	X															
0	0	0															
0	1	1															
1	0	1															
1	1	1															

FIGURE 4.3 Representations of an OR gate

XOR Gate

An XOR gate accepts two input signals

If both are the same, the output is 0; otherwise, the output is 1

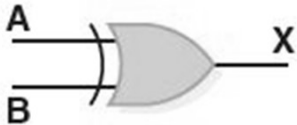
Boolean Expression	Logic Diagram Symbol	Truth Table															
$X = A \oplus B$		<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	X	0	0	0	0	1	1	1	0	1	1	1	0
A	B	X															
0	0	0															
0	1	1															
1	0	1															
1	1	0															

FIGURE 4.4 Representations of an XOR gate

XOR Gate

Note the difference between the XOR gate and the OR gate; they differ only in one input situation

When both input signals are 1, the OR gate produces a 1 and the XOR produces a 0

XOR is called the *exclusive OR* because its output is 1 if (and only if):

- *either* one input *or* the other is 1,
- *excluding* the case that they both are

NAND Gate

The NAND (“NOT of AND”) gate accepts two input signals

If both are 1, the output is 0; otherwise, the output is 1

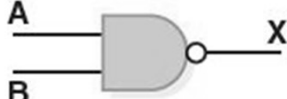
Boolean Expression	Logic Diagram Symbol	Truth Table															
$X = (A \cdot B)'$		<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	X	0	0	1	0	1	1	1	0	1	1	1	0
A	B	X															
0	0	1															
0	1	1															
1	0	1															
1	1	0															

FIGURE 4.5 Representations of a NAND gate

NOR Gate

The NOR (“NOT of OR”) gate accepts two inputs. If both are 0, the output is 1; otherwise, the output is 0.

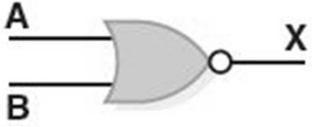
Boolean Expression	Logic Diagram Symbol	Truth Table															
$X = (A + B)'$		<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	X	0	0	1	0	1	0	1	0	0	1	1	0
A	B	X															
0	0	1															
0	1	0															
1	0	0															
1	1	0															

FIGURE 4.6 Representations of a NOR gate

AND Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

OR Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

XOR Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

NAND Truth Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

NOR Truth Table

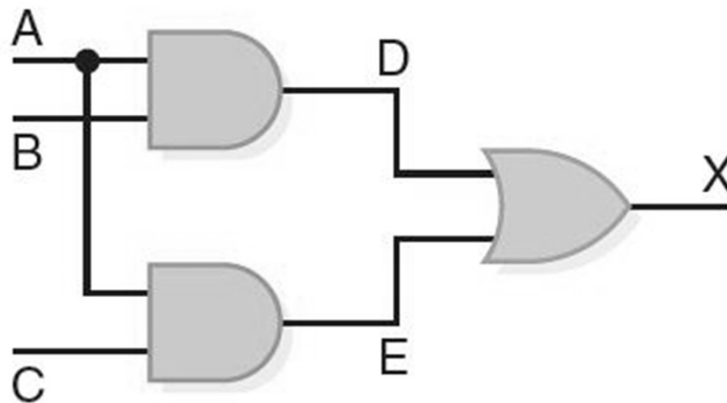
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

Review of Gate Processing

Gate	Behavior
NOT	Inverts its single input
AND	Produces 1 if all input values are 1
OR	Produces 0 if all input values are 0
XOR	Produces 0 if both input values are the same
NAND	Produces 0 if all input values are 1
NOR	Produces 1 if all input values are 0

Combinational Circuits

Gates are combined into circuits by using the output of one gate as the input for another



**This same
circuit using a
Boolean
expression is
 $AB + AC$**

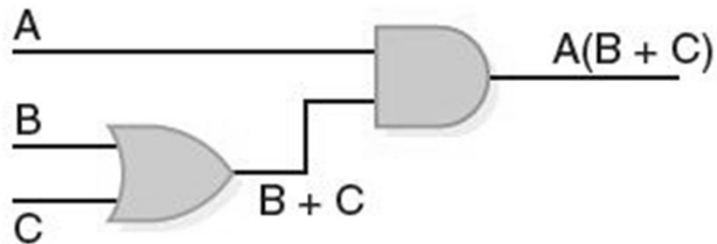
Combinational Circuits

A	B	C	D	E	X
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

Three inputs require eight rows to describe all possible input combinations

Combinational Circuits

Consider the following Boolean expression $A(B + C)$



A	B	C	$B + C$	$A(B + C)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Does this truth table look familiar?

Compare it with previous table